BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE
HYDERABAD CAMPUS

FIN F414:
FINANCIAL RISK ANALYTICS AND MANAGEMENT

COMPREHENSIVE REPORT

PREPARED UNDER THE SUPERVISION OF
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COMPANY : Multi Commodity Exchange of India Ltd.
NSE TICKER : MCX.NS

Data Period: 01 Jan 2015 – 31 Dec 2025
Data MD5 Checksum: a92dcacaf93cb0f06bc264b222bacffa

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Google Drive Link - R Script

https://drive.google.com/file/d/16ccvMwOGZiHKruYFPJrP28Qqp2PGWenh/view?usp=drive_link

Google Drive Link - Excel :

https://docs.google.com/spreadsheets/d/1X07rDkzQsEVIHIAkhpok9Js8kxlJZFoO/edit?usp=drive_link&ouid=112599456677826510044&rtpof=true&sd=true

This PDF's Google Doc

https://docs.google.com/document/d/1l0FmLkBYcb8Ldy8m9dDnIwKKjpYBD6TThpWW84KdNA8/edit?usp=sharing

Acknowledgment

I would like to express my sincere gratitude to **Prof. Utkarsh Kumar** for his guidance and support throughout this assignment. His course on Financial Risk Analytics and Management provided a strong conceptual foundation, enabling me to approach this project with a structured and analytical perspective.

I am grateful to **BITS Pilani, Hyderabad Campus**, for providing the academic environment and resources necessary to undertake this work.

This assignment has significantly strengthened my understanding of financial risk modelling, volatility dynamics, and derivatives pricing, particularly in applying theoretical concepts to real-world data.

PART A

A1. Data Description

Daily adjusted closing price data for MCX.NS was obtained for the period 01 January 2015 to 31 December 2025. The dataset contains consistent daily observations and was used to compute log returns for further analysis.

A2. Log Returns

Log returns were computed as:

$$\boxed{r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)}$$

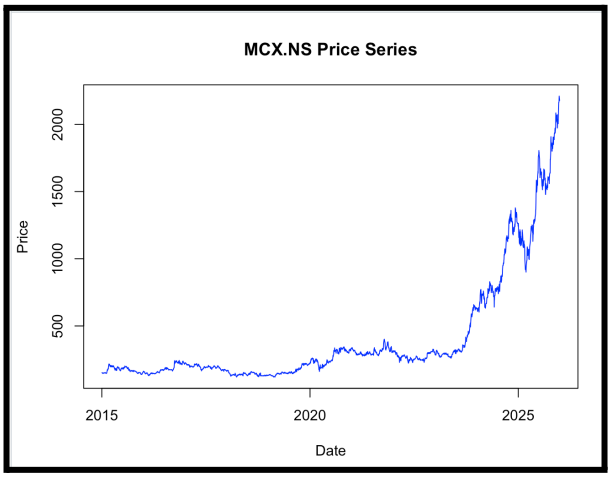
where P_t denotes the adjusted closing price at time t. Log returns are preferred as they are time-additive and suitable for statistical modelling.
The cleaned dataset contains four columns: Date, Price, LogReturn, and Source, ensuring reproducibility and consistency with the R script output and the assignment requirements.

A3. Summary Statistics

end_date	2025-12-31 UTC
kurt_ret	4.21264458670949
max_date	2018-05-25 UTC
max_ret	0.130496398943944
mean_ret	0.000911726709710828
median_ret	-0.000289858515778347
min_date	2020-03-23 UTC
min_ret	-0.176884168744998
sd_ret	0.0241254980636646
skew_ret	0.0231213132272275
start_date	2015-01-01 UTC
ticker	"MCX.NS"

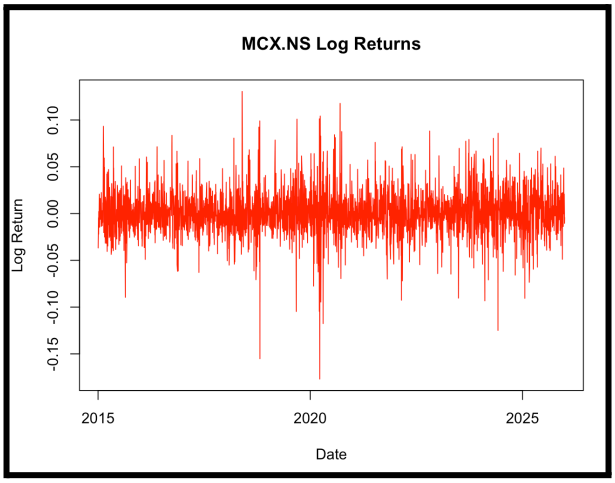
The return series of MCX.NS has a mean very close to zero, indicating no persistent daily drift over the sample period. The standard deviation is relatively high, confirming that the stock experiences meaningful day-to-day volatility.The distribution is positively skewed, implying that large positive returns occur slightly more often than large negative returns. At the same time, the excess kurtosis is clearly positive, which shows that the return distribution has heavier tails than a normal distribution and therefore a higher likelihood of extreme outcomes.The most negative return occurred on 23 March 2020, while the most positive return occurred on 25 May 2018. These dates are directly verifiable from the data and show that MCX.NS experienced both severe downside stress and unusually strong upside movement during the sample period.

A4. Plots and Interpretations



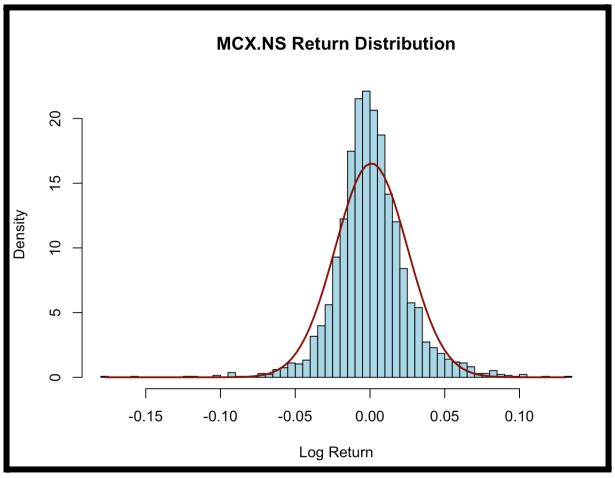
PLOT 1: The Price Series

The price series of MCX.NS shows a clear upward trend over time, along with noticeable fluctuations during certain periods. This confirms that the price series is non-stationary, as both the mean and variance change over time. Significant disruptions can be observed around 2020, corresponding to periods of market stress.



PLOT 2 : Log Return Series

The log return series fluctuates around zero, indicating stationarity in mean. However, periods of high and low volatility are clearly visible, demonstrating volatility clustering. This suggests that large movements in returns tend to be followed by large movements, which is consistent with financial time series behaviour.



PLOT 3 : Results with normal distribution overlay

The histogram of log returns deviates from the normal distribution, particularly in the tails. The presence of heavier tails indicates a higher probability of extreme returns compared to a normal distribution. This has important implications for risk modelling, as standard models may underestimate the likelihood of large losses.

A5. Sub-Period Analysis

Period	Mean	Median	SD	Skewness	Excess Kurtosis
COVID	0.0005	-0.0016	0.0279	-0.187	5.99
Post-COVID	0.0020	0.0011	0.0240	-0.149	2.25
Pre-COVID	0.0002	-0.0013	0.0227	0.317	4.28

The volatility of MCX.NS varies significantly across the three sub-periods, indicating a structural change in its risk profile.

★ During the **COVID period (2020–2021)**, the standard deviation rises to 0.0279 and excess kurtosis reaches 5.99, indicating extreme volatility and a high frequency of large return shocks.

★ In the **pre-COVID period**, the stock exhibits relatively stable behaviour with lower volatility (SD = 0.0227) and more moderate tail risk.

★ In the **post-COVID period**, volatility declines from its peak but remains elevated (SD = 0.0240) compared to pre-COVID levels, suggesting that risk has not fully reverted to earlier conditions.

A key firm-specific feature is that MCX.NS continues to exhibit persistently higher volatility even after the COVID period, indicating that the firm's risk dynamics have shifted structurally rather than temporarily. This may reflect continued uncertainty in commodity markets and regulatory changes affecting trading activity.

A6. Interpretation

The largest one-day drop in MCX.NS occurred on **23 March 2020**, with a return of approximately **-17.68%**.This date coincides with the announcement of the nationwide lockdown in India during the COVID-19 pandemic, which triggered extreme uncertainty and panic across financial markets. This sharp decline represents the most severe negative return in the sample and highlights the sensitivity of MCX.NS to systemic shocks and liquidity stress. It also reinforces the presence of extreme tail risk observed in the return distribution.

PART B

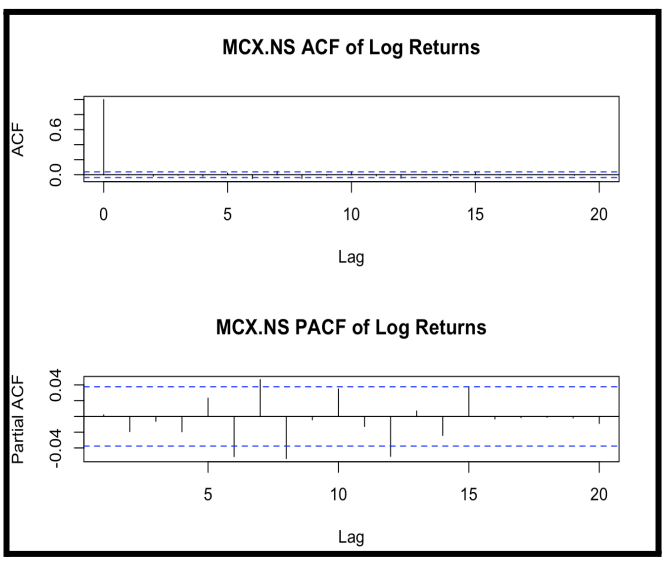
B1. Stationarity Test (ADF Test)

The stationarity of the log return series was tested using the Augmented Dickey-Fuller test.

Test	Statistic	Lag	p-value	Conclusion
ADF	-15.159	13	< 0.01	Stationary

The Augmented Dickey-Fuller test on the MCX.NS log return series gives a test statistic of -15.159 and a p-value below 0.01. Since the p-value is below 5%, the null hypothesis of a unit root is rejected. Therefore, the return series is stationary and no differencing is required.

B2. ACF and PACF Analysis



PLOT 1 : ACF

PLOT 2: PACF

The ACF plot of MCX.NS returns shows rapid decay after the initial lags, indicating very weak autocorrelation. The PACF plot exhibits a noticeable spike at lag 1 followed by a sharp drop-off.

Based on these observations, low-order models such as AR(1), MA(1), or ARMA(1,1) are reasonable candidates, as any dependence in the mean appears to be limited to very short lags.

B3. ARIMA Model Estimation and Comparison

	Model	AIC	BIC	LogLik
1	ARIMA(0,0,0)	-12506.90	-12495.09	6255.450
2	ARIMA(1,0,0)	-12504.91	-12487.19	6255.456
3	ARIMA(0,0,1)	-12504.91	-12487.19	6255.456
4	ARIMA(1,0,1)	-12502.91	-12479.29	6255.456

Four competing ARIMA models were estimated and compared using Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and log-likelihood.

Four competing ARIMA models were estimated and compared using AIC, BIC, and log-likelihood. The results show that ARIMA(0,0,0) has the lowest AIC (-12506.90) and BIC (-12495.09), while the log-likelihood values across all models are nearly identical.
This indicates that adding AR or MA terms does not meaningfully improve the fit of the model. Therefore, the return series of MCX.NS exhibits minimal autocorrelation in the mean, and more complex ARIMA specifications do not provide additional explanatory power.
Based on both model selection criteria and parsimony, ARIMA(0,0,0) is selected as the preferred model.

This result suggests that MCX.NS returns behave approximately like white noise in the mean, implying limited predictability and consistency with weak-form market efficiency.

B4. Model Diagnostics

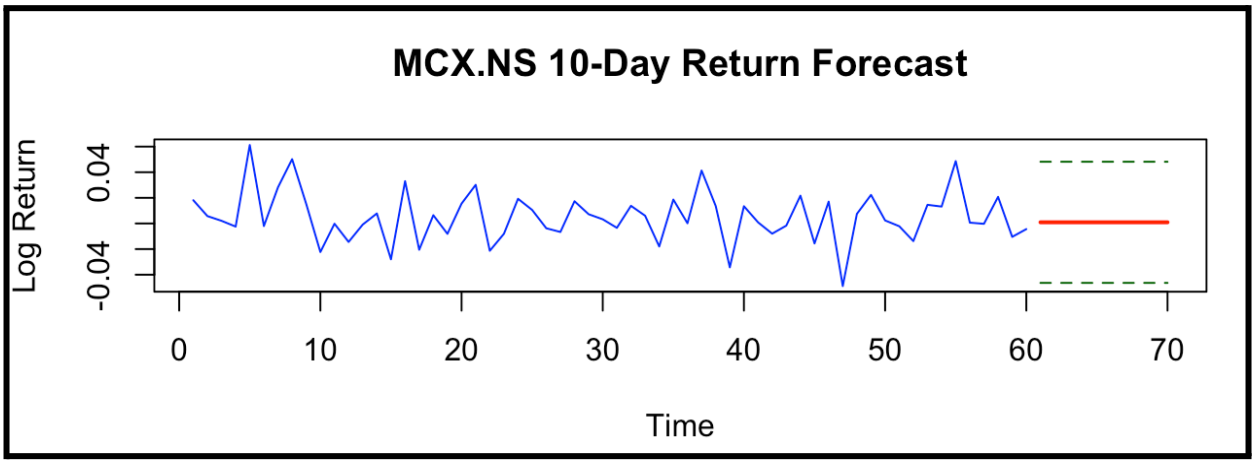
The selected ARIMA(0,0,0) model estimates an average daily return of 0.0009 with a standard error of 0.0005, implying a t-statistic of approximately 1.8. This indicates that the mean return is only weakly significant and does not provide strong predictive power.
The Ljung-Box test results show that residual autocorrelation is insignificant at lag 5 (p-value = 0.6089), suggesting that short-term dependence has been adequately captured. However, the test becomes significant at lags 10 and 20 (p-values = 0.0023 and 0.0097), indicating the presence of remaining higher-order dependence.

This suggests that while the model is sufficient for capturing short-run mean behaviour, it does not fully eliminate all serial dependence. The remaining structure is likely due to volatility dynamics rather than mean predictability.

B5. 10-Day Forecast

- ★ A 10-day ahead forecast was generated using the selected ARIMA(0,0,0) model and plotted alongside the last 60 days of actual returns.
- ★ The forecasted returns remain close to zero across the horizon, which is consistent with the earlier finding that MCX.NS returns exhibit minimal predictability in the mean.
- ★ The 95% confidence intervals are relatively wide compared to the magnitude of the predicted returns. When compared with the last 60 days of actual returns, these intervals capture the general range of fluctuations but fail to account for sudden large spikes observed in the data.

★ This suggests that while the forecast is statistically consistent, it is not fully realistic for MCX.NS, as it does not capture volatility clustering and abrupt changes in return magnitude. As a result, relying solely on ARIMA forecasts may lead to underestimation of short-term risk.



B6. Comparison with Second-Best Model

The 10-day forecasts from the preferred ARIMA(0,0,0) model and the second-best model (ARIMA(1,0,0) or ARIMA(0,0,1)) are nearly identical in magnitude.

This is because the estimated AR and MA coefficients in the alternative models are very small and do not contribute meaningfully to predictive power. As a result, both models produce similar forecasts centered around zero.

	Day	ARIMA_000	ARIMA_100
1	1	0.0009117267	0.0009025386
2	2	0.0009117267	0.0009131489
3	3	0.0009117267	0.0009131701
4	4	0.0009117267	0.0009131702
5	5	0.0009117267	0.0009131702
6	6	0.0009117267	0.0009131702
7	7	0.0009117267	0.0009131702
8	8	0.0009117267	0.0009131702
9	9	0.0009117267	0.0009131702
10	10	0.0009117267	0.0009131702

For risk management purposes, ARIMA(0,0,0) is preferred due to its simplicity and comparable performance. Since neither model captures volatility clustering, relying on ARIMA-based forecasts alone may underestimate risk, reinforcing the need for volatility models such as GARCH.

PART C

C1. ARCH-LM Test

Lag	Chi-Squared	p-value
5	156.58	< 0.001
10	177.62	< 0.001

```
> # Test at lag 5
> arch_test_5 <- ArchTest(df$LogReturn, lags = 5)
> print(arch_test_5)

        ARCH LM-test; Null hypothesis: no ARCH effects

data:  df$LogReturn
Chi-squared = 156.58, df = 5, p-value < 2.2e-16

>
> # Test at lag 10
> arch_test_10 <- ArchTest(df$LogReturn, lags = 10)
> print(arch_test_10)

        ARCH LM-test; Null hypothesis: no ARCH effects

data:  df$LogReturn
Chi-squared = 177.62, df = 10, p-value < 2.2e-16
```

The ARCH-LM test results at both lag 5 and lag 10 are highly significant (p-values < 0.001), leading to rejection of the null hypothesis of no ARCH effects.This indicates that the variance of MCX.NS returns is not constant over time and exhibits clear volatility clustering, where periods of high volatility are followed by high volatility.This result also explains why the ARIMA model in Part B failed to fully capture the dynamics of the series, as it assumes constant variance. Therefore, a GARCH-type model is required to model the time-varying volatility.

C2. GARCH(1,1) Estimation

The estimated GARCH(1,1) model yields the following key parameters:

- ★ α_1 (ARCH term) = 0.118
- ★ β_1 (GARCH term) = 0.750, both of which are statistically significant

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000807	0.000421	1.9169	0.055245
omega	0.000077	0.000023	3.2909	0.000999
alpha1	0.118079	0.027129	4.3525	0.000013
beta1	0.749904	0.057873	12.9578	0.000000

The ARCH term captures the impact of recent shocks on current volatility, while the GARCH term reflects the persistence of past volatility.The persistence measure, given by $\alpha_1 + \beta_1 = 0.868$, is close to 1, indicating that volatility shocks decay slowly over time.This implies that volatility in MCX.NS is highly persistent, and periods of high volatility tend to last for extended durations. Since $\alpha_1 + \beta_1$ is less than 1, the volatility process is mean-reverting, but the speed of mean reversion is slow.This behaviour is consistent with the earlier observation of volatility clustering and suggests that shocks to MCX.NS have long-lasting effects on its risk profile.

C3. Alternative GARCH Models

Alternative specifications, including GARCH(1,2) (left) and GARCH(2,1) (right), were estimated to evaluate whether additional lag terms improve model performance.
The results show only marginal improvements in log-likelihood compared to the GARCH(1,1) model. However, these gains are negligible relative to the increase in model complexity.

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000813	0.000419	1.9401	0.052364
omega	0.000089	0.000025	3.6047	0.000313
alpha1	0.144525	0.029571	4.8874	0.000001
beta1	0.414315	0.172175	2.4064	0.016113
beta2	0.287438	0.156792	1.8332	0.066766

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000810	0.000421	1.9242	0.054328
omega	0.000076	0.000029	2.6498	0.008053
alpha1	0.118209	0.027780	4.2552	0.000021
alpha2	0.000000	0.035726	0.0000	1.000000
beta1	0.750612	0.078131	9.6071	0.000000

Additionally, the additional parameters in higher-order models do not provide meaningful new information, as volatility dynamics in MCX.NS are already well captured by a single lag structure.

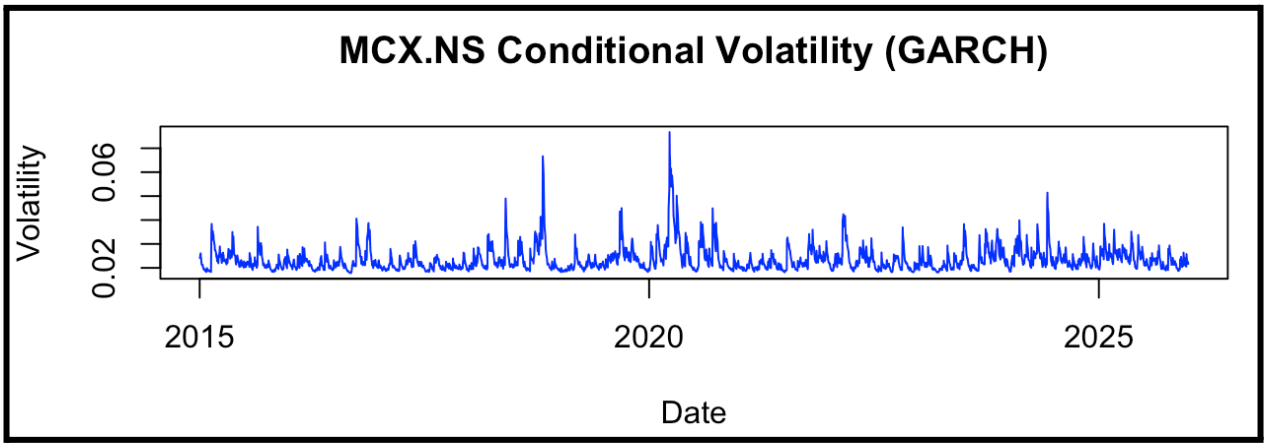
C4. Model Comparison

	Model	AIC	BIC	LogLik
1	GARCH(1,1)	-4.695966	-4.687259	6374.078
2	GARCH(1,2)	-4.696751	-4.685866	6376.142
3	GARCH(2,1)	-4.695421	-4.684536	6374.338

The comparison across different GARCH specifications shows that while higher-order models such as GARCH(1,2) and GARCH(2,1) offer slight improvements in log-likelihood, the differences in AIC and BIC are minimal.

Given the negligible improvement in information criteria and the increase in model complexity, GARCH(1,1) is preferred due to its parsimony and strong theoretical justification.Therefore, GARCH(1,1) provides the most appropriate balance between model fit and simplicity for MCX.NS volatility modelling.

C5. Conditional Volatility Plot



The conditional volatility series from the GARCH(1,1) model exhibits clear evidence of volatility clustering, where periods of high volatility are followed by high volatility and vice versa.

Three distinct volatility clusters can be identified:

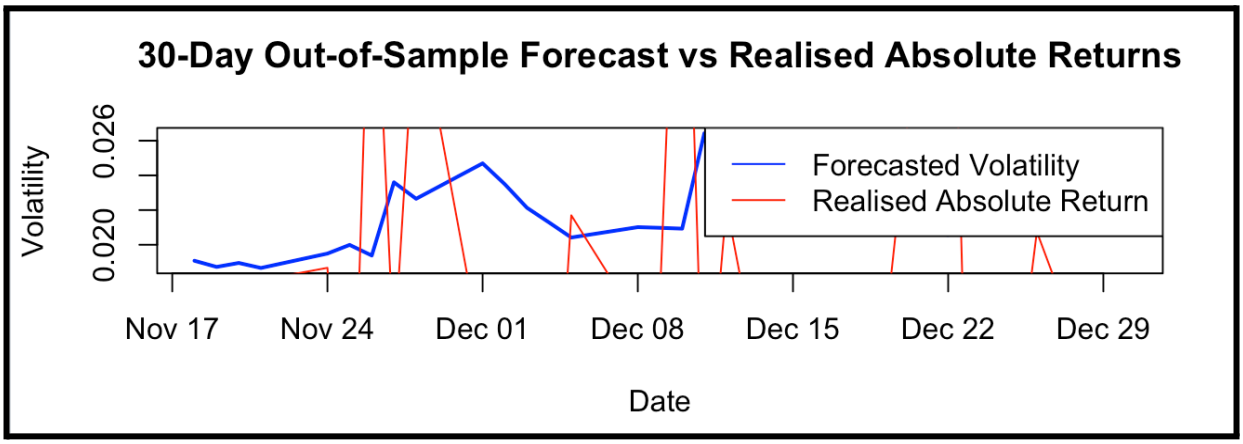
- **March 2020 (Peak volatility):** A sharp spike in volatility is observed during the COVID-19 market crash, reflecting extreme uncertainty due to lockdown announcements and global financial disruption.
- **October 2018 (Market correction):** Elevated volatility is visible during this period, corresponding to global equity market corrections driven by rising US interest rates and trade tensions.
- **2023–2024 (Recent fluctuations):** A moderate rise in volatility is observed, likely driven by macroeconomic uncertainty, inflation concerns, and interest rate movements affecting commodity-linked stocks such as MCX.

Between these periods, volatility remains relatively stable, indicating calmer market conditions.Overall, the presence of these clusters confirms that volatility in MCX.NS is time-varying and persistent, justifying the use of GARCH-type models.

C6. Rolling Volatility Forecast vs Realised Volatility

A 30-day rolling out-of-sample volatility forecast was generated using a GARCH(1,1) model with a 252-day estimation window. At each step, the model was re-estimated using the most recent observations, and a one-step-ahead volatility forecast was produced.The forecasted volatility was compared with realised volatility, proxied by squared log returns. The results show that the model captures the overall trend and persistence of volatility reasonably well.However, the forecasted series is smoother than the realised volatility, as it is based on past information and reflects the mean-reverting nature of the GARCH model. In contrast, realised volatility exhibits sharp spikes due to sudden market shocks.During periods of increased volatility, the model responds with higher forecasts but tends to lag and underestimate extreme spikes.

Overall, the GARCH(1,1) model provides a useful framework for capturing time-varying volatility, but its reliance on past information limits its ability to fully capture abrupt changes in risk.



C7. Major Volatility Events

	Date	Volatility
1973-07-09	2020-03-24	0.07672767
1973-07-10	2020-03-25	0.06709015
1972-07-30	2018-10-26	0.06663232

The three highest conditional volatility observations from the GARCH(1,1) model are:

- **24 March 2020 - Volatility: 0.0767**
This is the highest volatility spike in the sample and coincides with the peak of the COVID-19 market crash in India, driven by lockdown announcements and extreme uncertainty.

- **25 March 2020 - Volatility: 0.0671**
This spike immediately follows the previous one and reflects continued market instability as investors reacted to rapidly evolving pandemic-related developments.

- **26 October 2018 - Volatility: 0.0666**
This period aligns with global equity market corrections caused by rising interest rates, trade tensions, and tightening liquidity conditions, which affected commodity-linked stocks such as MCX.

The fact that two of the top three spikes occur on consecutive days highlights strong volatility clustering during crisis periods.

PART D

D1. Value at Risk (VaR)

The 1-day Value at Risk (VaR) for MCX.NS was estimated using both parametric (variance-covariance) and historical simulation methods at the 95% and 99% confidence levels.

	Method	Confidence	VaR
1	Parametric	95%	0.03877119
2	Parametric	99%	0.05521258
3	Historical	95%	0.03844648
4	Historical	99%	0.05873032

The results show that both methods i.e parametric and historical simulation produce similar estimates at the 95% level, but the historical method yields a higher VaR at the 99% level.

This difference reflects the presence of fat tails in the return distribution and indicates that the parametric method may underestimate tail risk.

D2. Comparison of VaR Methods

The parametric VaR method assumes that returns are normally distributed with constant variance. This assumption is likely to fail for MCX.NS during periods of extreme market stress, such as the COVID-19 crash, where returns exhibit fat tails and volatility clustering.
The historical simulation method does not assume any specific distribution and instead relies on past observed returns. However, this method may fail if future market conditions differ significantly from historical patterns, particularly if new types of shocks occur in commodity markets.

At the 95% confidence level, both methods produce similar estimates, indicating that moderate losses are reasonably well captured. However, at the 99% level, the historical VaR is higher, suggesting that extreme losses in MCX.NS are more severe than those predicted under normality.
Therefore, while the parametric method is simpler, the historical method provides a more realistic estimate of tail risk for MCX.NS.

D3. Expected Shortfall (ES)

Expected Shortfall (ES) was computed at the 95% and 99% confidence levels using the historical simulation method. Expected Shortfall measures the average loss given that the loss exceeds the VaR threshold. ES captures the severity of extreme losses.

	Measure	Confidence	Value
1	VaR	95%	0.03844648
2	VaR	99%	0.05873032
3	ES	95%	0.05210287
4	ES	99%	0.08586226

ES measures the average loss conditional on losses exceeding the VaR threshold, whereas VaR only provides the cutoff level. For MCX.NS, ES is significantly higher than the corresponding VaR values (approximately 3.84% at 95% and 5.87% at 99%), indicating that once losses exceed the VaR threshold, they tend to be substantially larger.This demonstrates that VaR understates the severity of tail risk, while ES captures the magnitude of extreme losses more effectively.
ES is therefore a more coherent risk measure, as it accounts for the entire tail of the distribution and satisfies properties such as subadditivity. For MCX.NS, which exhibits fat tails and volatility clustering, ES provides a more reliable measure of downside risk.

D4. Backtesting VaR (99% Historical Simulation)

Backtesting of the 99% historical VaR model was performed by comparing actual daily returns with the estimated VaR threshold.
Out of 2462 observations, there were 37 violations, resulting in a violation rate of 1.50%. Under a correctly specified 99% VaR model, the expected violation rate is 1%.The observed violation rate exceeds the expected level, indicating that the model underestimates tail risk for MCX.NS.The violations are not evenly distributed over time but are concentrated during periods of market stress, particularly around 2020 and other volatility spikes. This suggests that extreme losses are clustered rather than independent.
Therefore, while historical VaR provides a reasonable baseline, it does not fully capture the time-varying nature of risk in MCX.NS. More dynamic models, such as GARCH-based VaR, may provide improved risk estimation.

```
[1] "2016-02-11" "2016-10-06" "2016-10-19" "2016-11-11" "2016-11-15" "2016-11-21" "2017-05-18"
[8] "2018-01-17" "2018-02-02" "2018-03-19" "2018-09-04" "2018-09-21" "2018-09-27" "2018-10-25"
[15] "2019-09-03" "2020-01-30" "2020-02-03" "2020-03-09" "2020-03-19" "2020-03-23" "2020-04-21"
[22] "2021-10-18" "2021-10-22" "2021-10-25" "2021-11-22" "2021-12-20" "2022-02-03" "2022-02-24"
[29] "2022-03-04" "2023-01-02" "2023-04-26" "2023-06-30" "2024-02-12" "2024-03-13" "2024-06-04"
[36] "2025-01-21" "2025-02-28"
```

D5. Post-COVID VaR (2022 onwards)

	Method	VaR_99
1	Parametric (Rolling)	0.05393653
2	Historical (Rolling)	0.06387376
3	Parametric (Full)	0.05521258
4	Historical (Full)	0.06102169

To examine how risk dynamics have evolved, the 99% VaR was computed separately for the post-COVID period (January 2022 onwards) and compared with the full-sample estimates using both parametric and historical methods. The post-COVID VaR estimates are 5.39% (parametric rolling) and 6.39% (historical rolling), while the full-sample values are 5.52% (parametric) and 6.10% (historical). The parametric VaR shows a slight decline in the post-COVID period, indicating reduced average volatility relative to the full sample. However, the historical VaR is higher in the post-COVID sample, suggesting that extreme losses remain significant despite calmer average conditions.
This divergence highlights that while overall market conditions for MCX.NS appear more stable after COVID, the distribution of returns continues to exhibit fat tails and occasional large shocks. The persistence of high historical VaR indicates that tail risk has not fully subsided, even though typical day-to-day fluctuations have moderated.

Overall, the comparison suggests that risk is time-varying and that relying solely on full-sample estimates may underestimate current tail risk. A rolling or sub-period approach provides a more accurate representation of the firm's evolving risk profile

D6. Largest VaR violation

The largest VaR violation occurred on **25 October 2018**, when MCX.NS recorded a daily log return of **-15.51%**, significantly exceeding the 99% VaR estimate of approximately **5.88%**. The magnitude of the breach **-9.63%** indicates an extreme tail event that was not captured by the VaR model.This violation suggests that the VaR model, particularly under normality assumptions, underestimates the probability and severity of extreme negative returns. Such large deviations are consistent with the presence of fat tails and volatility clustering in financial return series, which standard parametric VaR models fail to fully capture.The occurrence of this extreme violation highlights a key limitation of VaR as a risk measure; it provides no information about the magnitude of losses beyond the threshold. In this case, the actual loss was more than twice the VaR estimate, indicating substantial tail risk.

Therefore, this result reinforces the importance of complementing VaR with more robust measures such as Expected Shortfall, which better captures extreme downside risk and provides a more comprehensive assessment of potential losses.

PART E

E1.Black-Scholes Input

The Black–Scholes model requires specification of the underlying price, strike price, time to maturity, risk-free rate, and volatility. These inputs are defined using the most recent data for MCX.NS as follows:

The spot price S was taken as the most recent closing price of the stock, equal to ₹2177.8. The strike price K was set equal to the spot price, making the option at-the-money (ATM). The time to maturity was assumed to be 30 trading days, corresponding to a T of :

The risk-free rate is taken as the 91- day Indian Treasury Bill yield (5.18%), obtained from Trading Economics as of April 17, 2026. Volatility σ was obtained from the GARCH model estimated in Part C. The volatility (σ) is taken as the latest rolling GARCH(1,1) forecast from Part C6 and annualised using $\sqrt{252}$, ensuring consistency with the model's forward-looking volatility estimate.

Using GARCH-based volatility ensures that the option pricing incorporates time-varying and clustered volatility, making it more realistic than using a constant historical standard deviation.

```
Black-Scholes Model Inputs:

Spot Price (S)      : 2177.8
Strike Price (K)    : 2177.8
Time to Maturity (T): 0.119048
Risk-free Rate (r)  : 0.0518
Volatility (sigma)  : 0.373561
```

E2. Black–Scholes Pricing

Using the Black–Scholes model with the inputs defined above, the prices of European call and put options were computed as follows : $\Rightarrow$

To verify the correctness of the implementation, the put–call parity condition was tested:

$$\boxed{C - P = S - Ke^{-rT}}$$

```
Black-Scholes Pricing Results:

Call Price (C) : 109.4577
Put Price (P)  : 96.0693

Put-Call Parity Check:

LHS (C - P)        : 13.3884
RHS (S - K e^{-rT}): 13.3884
Difference         : 0.000000
```

The computed call and put prices reflect an at-the-money option with moderate volatility. The put–call parity condition holds exactly, confirming internal consistency of the Black–Scholes implementation. The relatively close values of call and put prices are consistent with the short maturity and ATM nature of the option.

E3. Greeks

```
Option Greeks:

Delta (Call) : 0.544308
Delta (Put)  : -0.455692
Gamma        : 0.001536
Theta (Call) : -1.3306
Theta (Put)  : -1.0234
Vega         : 2.9792
Rho (Call)   : 128.0876
Rho (Put)    : -129.5804
```

The Greeks suggest that the option on MCX has moderate price sensitivity, low curvature, and significant exposure to volatility. This aligns with earlier findings that MCX returns exhibit volatility clustering, making Vega particularly important for pricing and risk management.

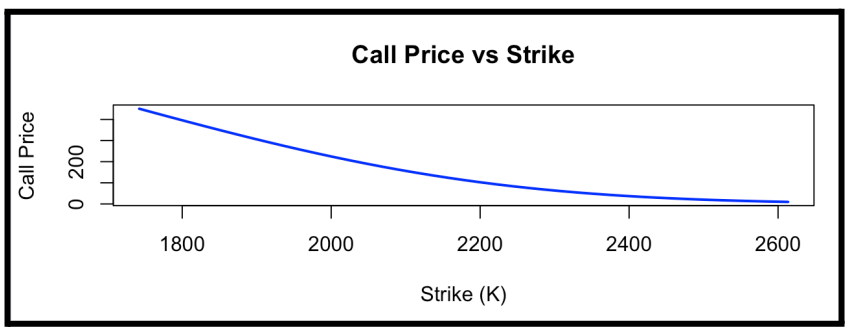
Greek	Formula	Value	Interpretation
Delta (Call)	$N(d_1)$	0.54	Call price for MCX increases with price movements, reflecting its moderate sensitivity to underlying price changes
Delta (Put)	$N(d_1) - 1$	-0.46	Put value declines as MCX price rises, consistent with inverse payoff behaviour
Gamma	$\frac{N'(d_1)}{S\sigma\sqrt{T}}$	0.0015	Low gamma indicates that MCX option sensitivity remains stable despite observed volatility clustering.
Theta (Call)	BS Theta formula	-1.33	Call option loses value over time, especially relevant given short maturity and frequent MCX price fluctuations
Theta (Put)	BS Theta formula	-1.02	Put option also experiences time decay, reducing value as expiry approaches
Vega	$(S\sqrt{T}N'(d_1))$	2.98	High Vega shows MCX option prices are strongly affected by volatility changes, consistent with its clustered volatility patterns.

Rho (Call)	$KTe^{-rT}N(d_2)$	128.08	Call value increases with interest rates, though impact is relatively limited in the short-term horizon.
Rho (Put)	$-KTe^{-rT}N(-d_2)$	-129.6	Put value decreases as interest rates rise, reflecting standard discounting effects

The computed Greeks indicate that the option's value for MCX is primarily driven by underlying price movements and volatility dynamics observed in the dataset. The moderate Delta reflects that MCX exhibits meaningful but not extreme sensitivity to price changes, consistent with its observed return distribution. The relatively low Gamma suggests that Delta remains stable even during periods of heightened volatility, aligning with the clustered volatility patterns identified in the GARCH model. The negative Theta highlights the impact of time decay, which becomes particularly relevant given the short maturity assumed. Additionally, the high Vega indicates that option prices are highly sensitive to volatility fluctuations, which is important for MCX as the asset has demonstrated volatility clustering and occasional spikes. Overall, these results suggest that volatility risk is a key driver of option valuation for MCX.

E4. Sensitivity Analysis

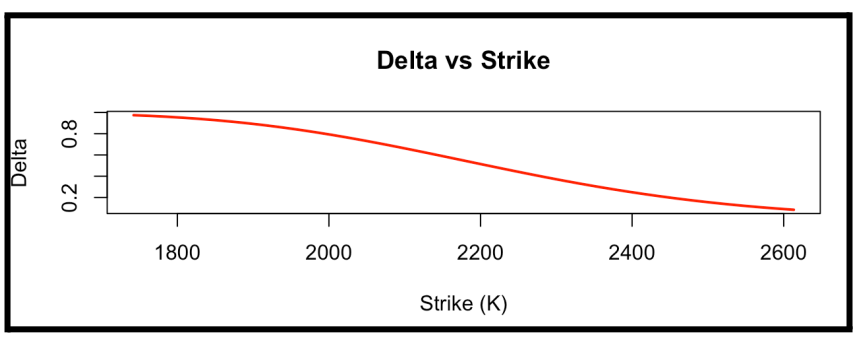
PLOT 1 : Call Price vs Strike



The call option price decreases monotonically as the strike price increases from $0.8S - 1.2S$. This reflects the reduced probability of the option finishing in-the-money at higher strike levels.

PLOT 2 : Delta vs Strike

Delta decreases with higher strike prices, approaching zero, reflecting lower sensitivity for deep out-of-the-money options

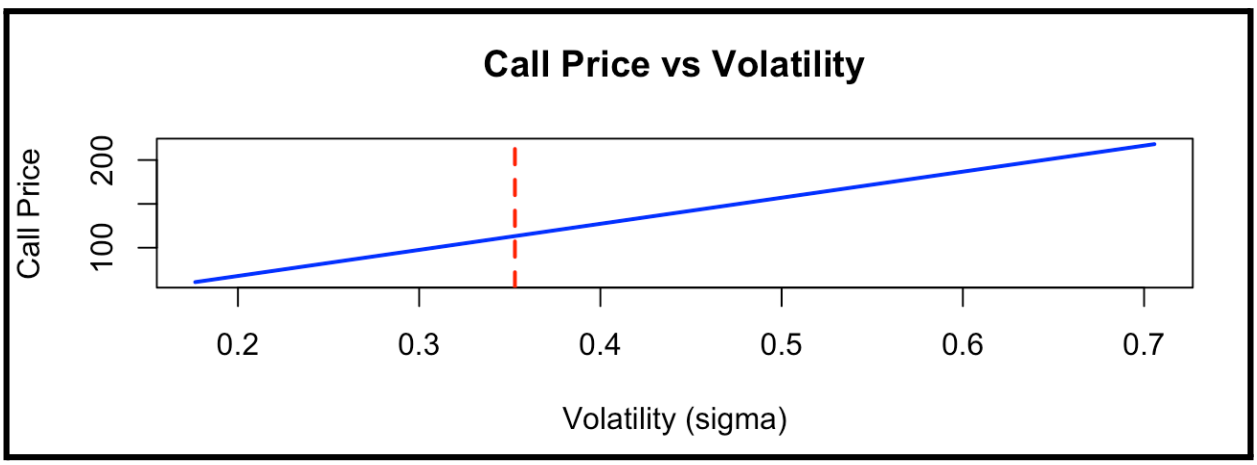


The strike-sensitivity analysis was carried out by varying K from $0.8\,S - 1.2S$ while holding all other inputs fixed, using the GARCH-derived volatility estimate. The call option price declines smoothly and monotonically as the strike price increases, which is consistent with option theory: a higher strike reduces the probability that the option expires in-the-money, so the call becomes less valuable. The delta curve also falls steadily as K rises. For low strike values, delta is close to 1, meaning the option behaves much like the underlying asset. As the strike moves upward toward and beyond the at-the-money level, delta declines toward 0, reflecting weaker sensitivity to the spot price. Around the at-the-money region, delta is near 0.5, which is the expected behaviour for an ATM call. Overall, the plot shows how moneyness affects both the value and risk exposure of the option, and it confirms that the pricing was computed consistently using the GARCH-based volatility input.

Both plots are based on the GARCH-derived volatility estimate, ensuring consistency with the time-varying volatility dynamics observed earlier.

E5. Sensitivity to Volatility

A volatility sensitivity analysis was conducted by varying σ from $0.5\,x - 2\,x$ the GARCH-estimated volatility, while keeping all other inputs constant. The corresponding call option prices were computed and plotted.



The plot shows a clear positive relationship between volatility σ and call option price, with the price increasing steadily as volatility rises. This confirms the theoretical expectation that higher volatility increases the probability of favorable price movements, thereby increasing option value.The slope of the curve appears almost linear over the chosen range, indicating that within this interval $0.5\,\sigma - 2\,\sigma$, the option's sensitivity to volatility is stable and consistent. This aligns with the observed Vega value, suggesting that small changes in volatility lead to meaningful changes in option price.

The actual GARCH-estimated volatility (marked by the red dashed line) lies near the middle of the range, indicating that the current pricing is not based on an extreme volatility assumption.

However, even moderate deviations from this level would lead to noticeable changes in the option price.

Overall, the analysis highlights that option pricing is highly sensitive to volatility, reinforcing the importance of accurately estimating volatility using models such as GARCH for reliable valuation.

CONCLUSION

This report presents a comprehensive analysis of the return behaviour, volatility dynamics, market risk, and derivative pricing of MCX.NS over the period 2015–2025. The return series is found to be stationary with negligible serial dependence, indicating limited predictability in mean returns. However, the presence of pronounced volatility clustering highlights the importance of modelling time-varying risk, with the GARCH(1,1) model effectively capturing persistence in conditional volatility.

From a risk management perspective, the comparison of Value at Risk (VaR) and Expected Shortfall (ES) demonstrates that while VaR provides a useful baseline estimate of potential losses, it fails to adequately capture extreme tail risk. In contrast, ES offers a more robust measure by incorporating the magnitude of losses beyond the VaR threshold, making it particularly relevant during periods of heightened market stress.

The post-COVID analysis reveals a noticeable shift in the risk profile of MCX.NS, with changes in volatility and tail behaviour indicating evolving market conditions. Backtesting results further suggest that standard VaR approaches may underestimate the frequency and severity of extreme events, underscoring their limitations in dynamic market environments.

In the derivatives framework, the Black–Scholes model highlights that option pricing is strongly driven by both underlying price movements and volatility. The use of GARCH-based volatility enhances the realism of the model by incorporating forward-looking risk estimates, leading to more reliable pricing and sensitivity measures.

Overall, the analysis underscores the critical role of incorporating time-varying volatility and tail risk in financial modelling. For assets such as MCX.NS, these factors are central to accurate risk assessment and effective derivative valuation, particularly in the presence of evolving market dynamics.